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Studierichting: Informatica
Oefeningenreeks 5G nr. 7.1

$$y = x^2, y = \frac{16}{x^2}, x \in [0,4] \text{en } y = 0$$

$$V_1 = \pi \int_0^2 x^4 dx = \pi \cdot \left[\frac{x^5}{5}\right]_0^2 = \frac{32\pi}{5}$$

$$V_2 = \pi \int_2^4 \frac{256}{x^4} dx = 256\pi \cdot \left[\frac{-1}{3x^3}\right]_2^4 = \frac{-256\pi}{192} + \frac{256\pi}{24}$$

$$V_1 + V_2 = \frac{32\pi}{5} - \frac{256\pi}{192} + \frac{256\pi}{24}$$

$$= \frac{32\pi}{5} - \frac{4\pi}{3} + \frac{32\pi}{3}$$

$$= \frac{32\pi}{5} + \frac{28\pi}{3} = \frac{96\pi}{15} + \frac{140\pi}{15}$$

$$= \frac{236\pi}{15}$$

References

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